



Ashish Pandey

Roll No.:244159002

M.Tech - Biomedical Science and Engineering

Medical Devices & Diagnostics

Indian Institute of Technology Guwahati

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Github | Portfolio

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology Guwahati	8.63 (Current)	2024-Present
B.Pharm	Dr. APJ Abdul Kalam Technical University, Lucknow	7.71	2020-2024
Senior Secondary	CBSE Board	70.2%	2017
Secondary	CBSE Board	10	2015

PROJECTS

- Design and Development of a Smart Spirometer with Integrated Patient Data Tracking** *July 2025 - Present*
Research Project under Prof. Raghvendra Gupta, IIT Guwahati
 - Designing and developing a portable smart spirometer with real-time respiratory data transmission and integrated patient data tracking.
 - Developing and training machine learning models for real-time computation of pulmonary function parameters (FEV1, FVC, FEV1/FVC ratio) to enable early detection of respiratory disorders.
- Machine Learning-Based Real Estate Price Prediction and Recommendation System** *June 2025 - July 2025*
Independent Project *Github*
 - Performed data preprocessing and exploratory data analysis (EDA) on 3,800 real estate records, ensuring high-quality, structured datasets for predictive analytics.
 - Developed, trained, and validated machine learning regression models for property price prediction, achieving an R2 score of 0.89 and enhancing model accuracy.
 - Implemented a content-based recommendation system using vectorization techniques and similarity algorithms to deliver personalized property suggestions.
- Solving Inventory Inefficiencies Using Advanced SQL Analytics** *May 2025 - June 2025*
Consulting and Analytics Club, IIT Guwahati *Github*
 - Analyzed large-scale inventory and sales datasets from 100+ retail stores using SQL, generating insights on stock levels, sales trends, and operational performance.
 - Designed and deployed interactive Power BI dashboards for real-time inventory tracking, demand forecasting, and performance monitoring to optimize warehouse operations.
 - Automated SQL queries and inventory workflows to minimize stockouts, streamline replenishment, and improve overall supply chain efficiency by 90%.

TECHNICAL SKILLS

- Programming & Databases:** Python, MySQL
- Data Analysis & Machine Learning:** Pandas, Numpy, Scikit-learn
- Data Visualization:** Matplotlib, Seaborn, Power BI, Excel

KEY COURSES TAKEN

- MOOCs:** MS Excel, SQL Bootcamp, Power BI for Data Analysis, Python Bootcamp, NumPy, Matplotlib & Pandas: Machine Learning, Statistics and Hypothesis Testing for Data science, Summer Analytics – Data Science (Certificate of Merit), Statistics and Hypothesis Testing for Data science

POSITIONS OF RESPONSIBILITY

- Core Team Member**, Sathi Counselling, IIT Guwahati *July 2025 - Present*
 - Coordinated 2+ counselling and wellness events, engaging 100+ students.
- Volunteer**, International Conference on Micro Nano Fluidics 2025, IIT Guwahati *July 2025 - Nov. 2025*
 - Led the hospitality and logistics team for ICOM 2025, managing accommodation, transportation, and on-site support to ensure a smooth delegate experience.

ACHIEVEMENTS

- Letter of Appreciation by HRD Minister Govt. of India**, Secured 10 CGPA, High School *July 2015*
- GATE XL**, Qualified GATE XL 2024 organized by IISC Bengaluru *Feb. 2024*

CONFERENCE

- Pandey, A., Gupta, R., Fuhrer, E., 2025.** A Framework for Point-of-Care Smart Spirometer with Integrated Machine Learning for Early Lung Disease Detection. Second International Conference on Micro Nano Fluidics (ICOM), IIT Guwahati, India.

EXTRACURRICULAR

- Event Coordinator**, Research Industrial Conclave 2025, IIT Guwahati *July 2025 - Oct. 2025*
- Student Intern**, RAA Internship an Initiative of Samagra Shiksha by Assam Govt. *2024, 2025 & 2026*
- Coordinator**, The New Nalanda, House 10, India International Science Festival 2024, IIT Guwahati *Dec. 2024*